



Introduction to Macroeconomics

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Problem Set 5

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Exercise 1

The short-run aggregate supply curve is given by :

$$Y_t = Y_n + \alpha(\pi_t - \pi_t^e)$$

where Y_t is the short-run GDP, Y_n is the long-run GDP, π_t is the observed inflation rate and π_t^e is the expected inflation rate. Let's assume that the Central Bank is able to control directly inflation (π_t) through the money supply and, therefore, it can surprise the agents with a level of inflation that is higher than what agents expect. What is the result of this policy in the short run? Explain why this result is incompatible with money neutrality in the short run.

While $Y_t = Y_n + \alpha(\pi_t - \pi_t^e)$ represents the short run aggregate supply (AS), $Y_t = Y_n$ represents the long-run aggregate supply (LRAS).

Let's assume that the Central Bank is able to control directly inflation (π_t) through money supply. Since $\alpha > 0$, if the agents underestimated the inflation rate ($\pi_t^e < \pi_t$), then $Y_t > Y_n$, that is, if the Central Bank "surprises" the agents with a higher inflation rate than expected, then it manages to push the short-run output value of that period (Y_t) above the natural level of output Y_n . However, if the agents overestimated the inflation rate ($\pi_t^e > \pi_t$), then $Y_t < Y_n$, which means that the output value of that period falls below the natural level of output. In this case, as the Central Bank manages to "surprise" agents with a higher inflation rate than they expected, the output on the short run (Y_t) is larger than the output on the long run or natural output (Y_n).

Money neutrality means that a change in money supply does not affect any of the real variables and that changes in money supply lead to equiproportional changes in the price level. In other words, if the Central Bank doubles the money supply, the price level doubles but there is no change in the real economy, and thus in the output. **In the short run**, if the Central Bank manages to increase the output Y_t above the natural level Y_n , **money is not neutral**. In the long run, the Central Bank cannot surprise agents at every period, who are rational and learn from the past. When the expectations of the agents are correct, we can see

that $\pi_t^e = \pi_t$, and thus $Y_t = Y_n$. Therefore, in the long run, we can assume that the agents anticipate correctly inflation, that $Y_t = Y_n$, and, as a result, money is neutral.

Exercise 2

Using the AS/AD model (the aggregate supply and aggregate demand model), please answer the following questions :

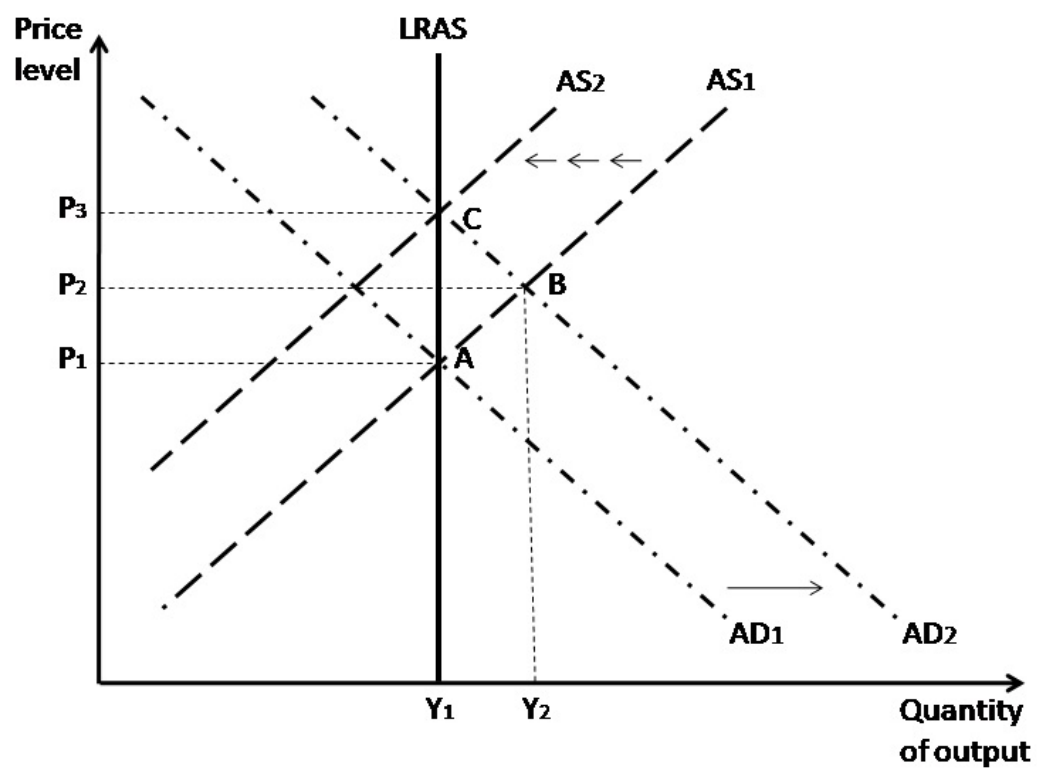
2.A The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, a wave of optimism overtakes the economy, and households start to consume more. Which one of the following statements is true (with respect to the initial situation, that is, point A) ?

- (a) In the short run, the output will increase and the price will decrease.
 - (b) In the long run, the output will be the same and the price will have increased.
 - (c) In the short run, only the price will increase.
 - (d) None of the above.
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Solution : (b)

A wave of optimism leads household to increase their consumption, which means that for any given price, the aggregate demand increases. Thus, AD shifts to the right (from AD_1 to AD_2). The economy moves thus from A to B : on the short run, price increases from P_1 to P_2 and output increases from Y_1 to Y_2 . Thus, (a) and (c) are false.

In the long run, as wages, prices and perceptions adjust, the AS shifts to the left so that point C, a new long-run equilibrium point, is reached. The economy moves thus from B to C. At C, like in any long run equilibrium, AS, AD and LRAS all cross. With respect to A, the output will be the same, but the price will have increased from P_1 to P_3 . Thus, (b) is correct.

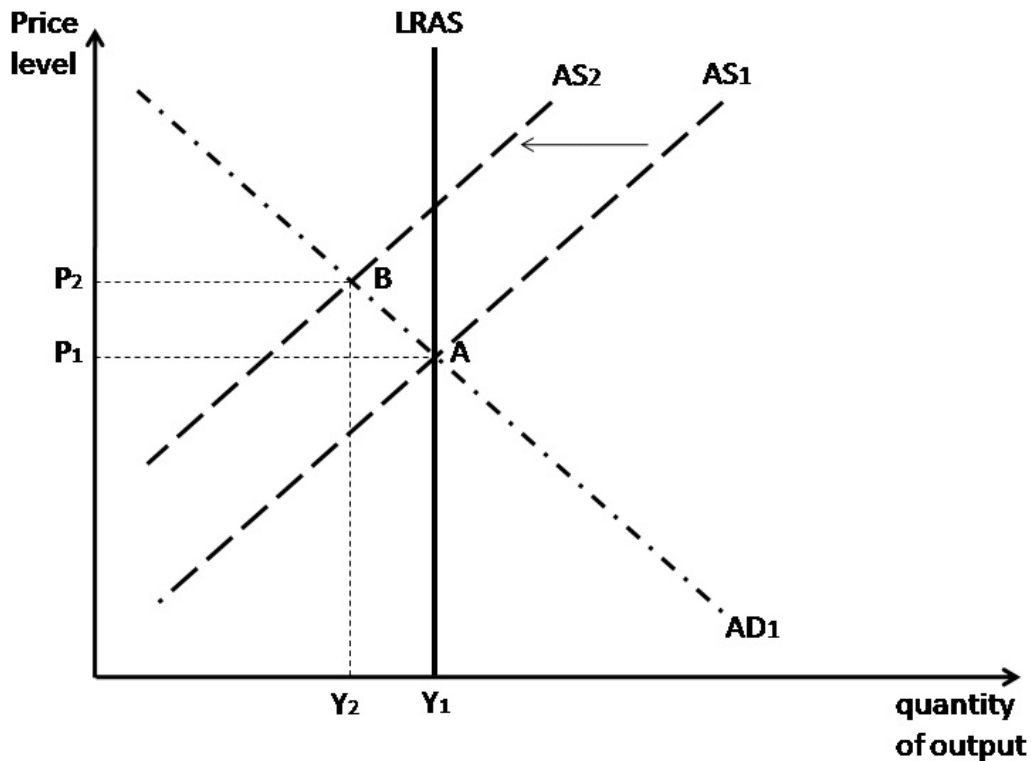


2.B The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, there is an increase in the price of oil. Which one of the following statements is true (with respect to the initial situation, that is, point A) ?

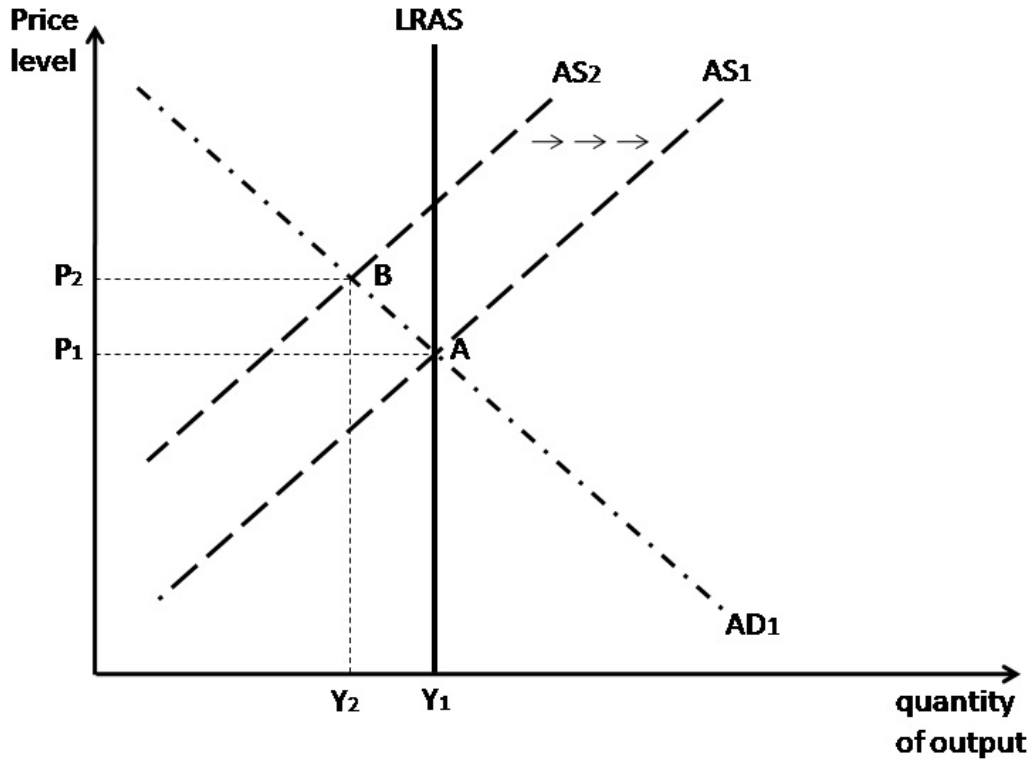
- (a) There is an episode of stagflation, in which output decreases and price increases. Once the former has decreased and the latter has increased, they will stay at their new values.
- (b) If the government wants the output to go back as soon as possible to its initial value, it must try to make the aggregate demand shift to the left.
- (c) If the government wants the output to go back as soon as possible to its initial value, it must try to make the aggregate demand shift to the right.
- (d) The government cannot do anything about stagflation, because the increase in the price of oil affects the aggregate supply and not the aggregate demand.

Solution : (c)

In the short run, the increase of the price of oil increases the cost of production of goods and, therefore, we have a shift of the AS to the left (from AS_1 to AS_2), which results in a decrease in output (from Y_1 to Y_2) and an increase in the price level (from P_1 to P_2). We have thus stagnation (falling output) and inflation (rising prices). Such a combination of events is called stagflation :

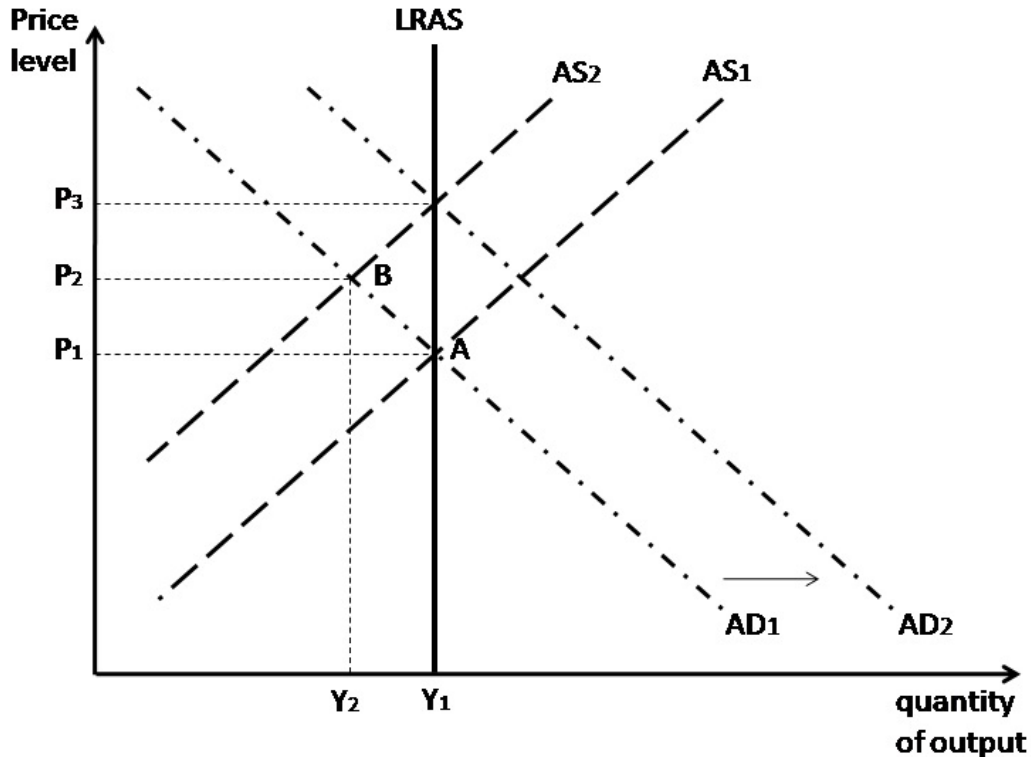


If there is no intervention of the government, in the long run, as wages, prices and perceptions adjust, the AS shifts to the right, and returns to its initial position :



Therefore, on the long run, the economy goes back to its initial state, A, where the price is Y_1 and the output is P_1 . Therefore, in the long run the output and the price will not remain at Y_2 and P_2 , respectively : (a) is false.

Now let's go back to the first graph of section 4.B., in which we had represented the short-run effect of the increase of the price of oil, that is, the onset of a period of stagflation. Let's assume that the government wants to avoid a period of stagflation and wants the output to go back at its initial value as soon as possible : it can achieve it by implementing a policy that makes the AD shift to the right from AD_1 to AD_2 , as shown in the following graph :



As a result, the output goes back to its initial value (Y_1) but the price increases even more than during the stagflation period, as it reaches $P_3 > P_1, P_2$. Therefore, (b) is false, (c) is correct and (d) is false.

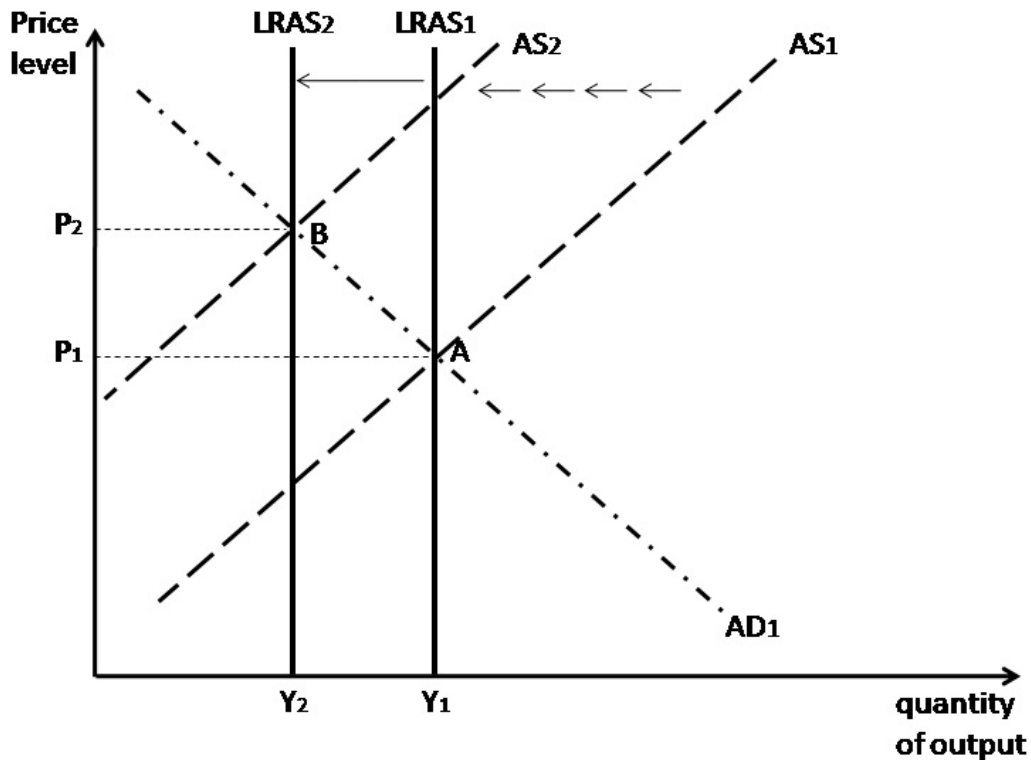
2.C The economy begins in a long-run equilibrium, represented by point A, where $Y = Y_1$ and $P = P_1$. Suddenly, part of the capital of the economy is **permanently** destroyed. Which one of the following statements is true on the long run (with respect to the initial situation, that is, point A) ?

- (a) In the long run, the output and the price decrease.
- (b) If the government makes the aggregate demand shift to the right, on the long run it can increase the output back again to its initial value.
- (c) If simultaneously there is a wave of pessimism and the households decide to cut back their consumption, the output decreases but we do not know whether the price level increases or decreases.
- (d) If simultaneously there is a wave of pessimism and the households decide to cut back their consumption, the output and the price decrease.

Solution : (c)

A permanent destruction of capital means that the long-run aggregate supply decreases, that is, LRAS shifts to the left, from LRAS₁ to LRAS₂. As the AD does not move, on the

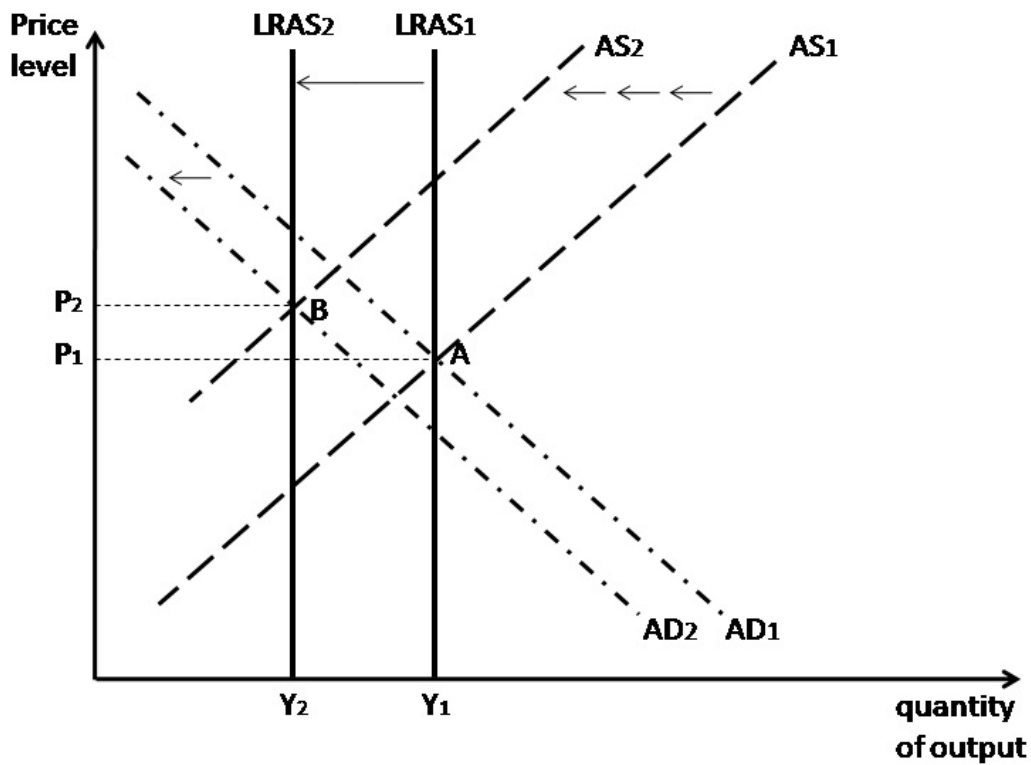
long run, the point B becomes the new equilibrium. How does that happen ? On the long run, as wages, prices and perceptions adjust, the AS shifts to the left until point B is reached.



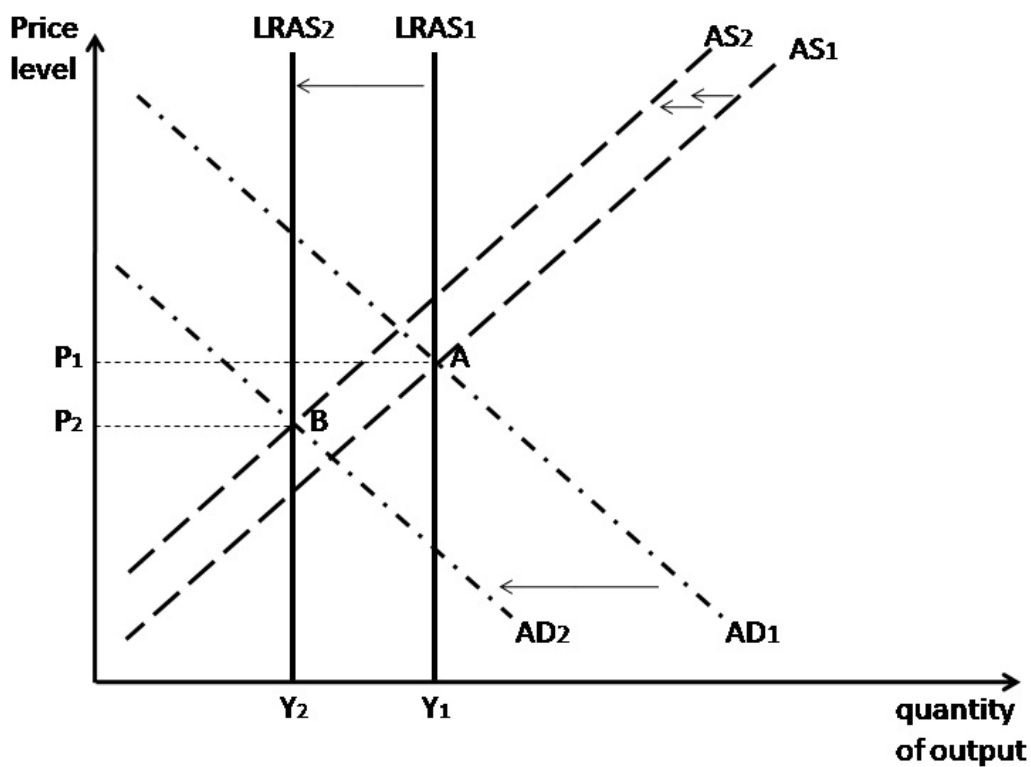
Thus, on the long run, output decreases to Y_2 and price increases to P_2 . This means that (a) is false.

We can also see from the previous graph that, if AD shifts to the right, on the long run the output will be necessarily Y_2 , and therefore a shift in the aggregate demand cannot take back the output to Y_1 on the long run. Thus, (b) is false.

But what happens if, as well as the shift of LRAS to the left, we have a shift of AD to the left due to the decrease in consumption of the households, which have become suddenly more pessimistic ? On the long run, the output will be at Y_2 . However, whether the price increases or decreases depends on the relative importance of the shift of the AD with respect to the shift of the LRAS. If the shift of the AD is relatively small, the price will increase on the long run ($P_2 > P_1$) :



However, if the shift of the AD is relatively large, the price will decrease in the long run ($P_2 > P_1$) :



Therefore, if we have shift of LRAS and AD to the left, on the long run the output decreases from Y_1 to Y_2 . However, we do not know what will happen to the price : whether

it increases or decreases will depend on the relative importance of the shift of the AD with respect to the shift of the LRAS. Thus, (c) is correct and (d) is false.

Exercise 3 (Exam June 2013)

Which of the following statements are correct ? :

- I. Expansionary monetary policy shifts the aggregate demand curve to the right and thus it increases income in the long run.
- II. If the marginal propensity to consume equals 0.6 and a country does not import anything, an increase of government expenditures of 1000 increases income by 2500.
- III. If the marginal propensity to consume equals 0.6 and the marginal propensity to import equals 0.1, then an increase of government expenditures of 1000 increases income by 2000.
- IV. Monetary and fiscal policy are automatic stabilizers, because they are used to stimulate production in times of recessions.

- (a) Only statement I is correct.
- (b) Statements I, II and III are correct and IV is wrong.
- (c) Statements II and III are correct and statements I and IV are wrong.
- (d) Only statement IV is correct.

Réponse (c)

I. Wrong : it is the aggregate demand in the short-run, not in the long-run.

II. Correct : $1/(1-mpc)=2.5$

III. Correct : $1/(1-mpc+mpi)=2$

IV. Wrong : by definition, economic stabilizers are not policies.

Exercise 4

After World War II, the stabilisation of the economy became an explicit objective of most governments in developed countries. Please answer the following questions :

1. According to some economists, an expansive fiscal policy in times of recession can, under some circumstances, be unable to increase output because consumers fail to increase their expenditures. Why ? Which are the underlying mechanisms ?
2. What are the economic stabilisers ? Why do we call them "stabilizers" and how do they work ?
3. Which are the short-run and long-run advantages and disadvantages of an expansive fiscal policy or an expansive monetary policy in times of recession ?

I. Fiscal policy consists in either an increase in public spending, a decrease in taxes, or both, which result in an increase in the consumers' income. The requirement for a fiscal policy to be effective and increase output in times of recession is that consumers use the extra income they receive to consume more, that is, the fiscal policy must lead to an increase in aggregate demand.

However, if the government increases public spending and/or decreases taxes, a budget deficit appears (or increases). In countries where the government cannot finance the budget

deficit through money creation, the only option available is to increase the government debt. Since government debt will have to be repaid in the future, it is likely that the government will end up collecting more taxes at some point in the future. According to the Ricardian equivalence proposition (also called the Barro-Ricardo equivalence theorem), consumers, who are rational and perfectly informed, anticipate the future increase in taxes and save all the extra income that results from the expansive fiscal policy. These savings will be used to pay for the extra taxes that will eventually result from the expansive fiscal policy. If the extra income is saved, aggregate demand fails to increase and the fiscal policy is completely ineffective.

However, the Ricardian equivalence proposition has been the object of numerous debates among economists. Indeed, for the Ricardian equivalence proposition to hold, the following conditions must be satisfied :

1. Individuals are rational.
2. Capital markets are perfect (households can borrow and save without constraints).
3. Households are infinitely-lived or behave as such (there are voluntary inter-generational transfers, since the households are altruistic towards future generations)
4. The government is perfectly credible (if there are any doubts whether the government is capable of honouring its debt, the interest rate would increase due to the risk premium).

II. The economic stabilizers are anticyclical mechanisms that allow the passive regulation of the economy by increasing (reducing) automatically output as soon as it moves downwards (upwards) :

(i) In times of recession, output goes down and, as a result, taxes on revenues also go down. When taxes go down, the income available to households increase and, as a result, consumption increases, pushing output upwards.

(ii) In times of recession, unemployment increases, the unemployment benefits increase. The consequence is an increase in the income available to household, which drives consumption (and output) upwards.

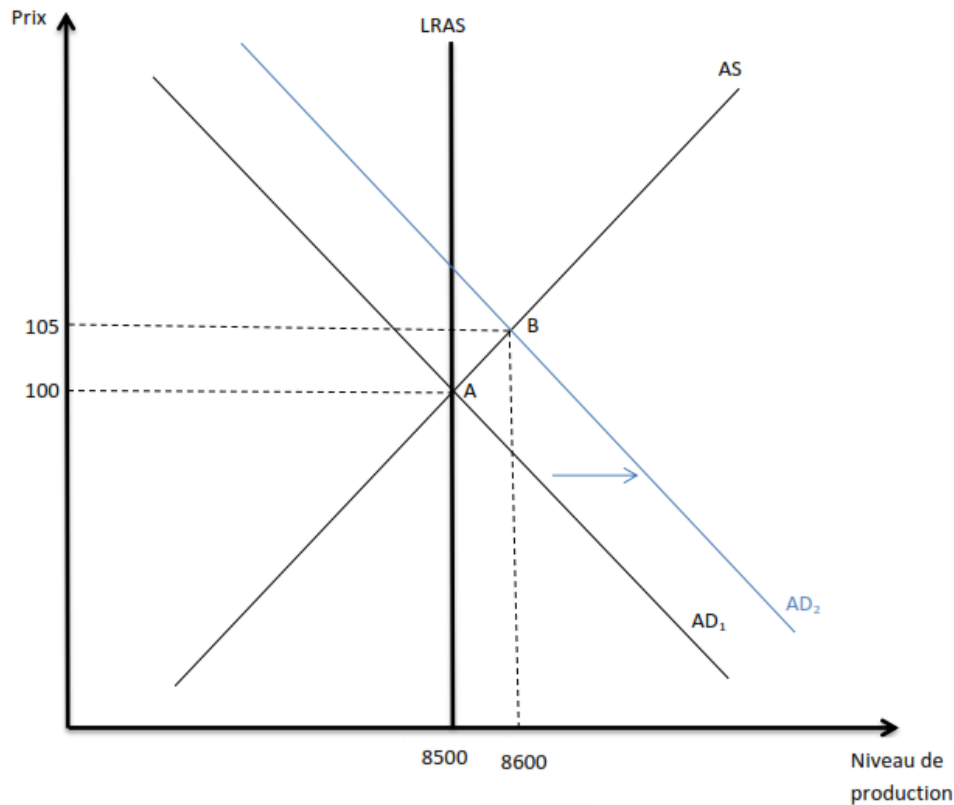
In times of boom, these same mechanisms reduce the income available to households, which drives consumption (and output) downwards. It should be noted that, in times of recession, both mechanisms generate a budget deficit (or increase the budget deficit).

III. In the short run, the main advantage of both is that they are able to increase output in times of recession. Since both stimulate demand, they can lead to an increase in the price of goods and service (inflation). Moreover, an expansive fiscal policy generally reduces the public savings and, as a result, the budget deficit increases.

In the long run, the budget deficit that results from fiscal policies, if too high, can become very serious. However, the inflation generated by expansive fiscal and/or monetary policies erodes the real value of debt. This, together with the increase in the output, makes it easier for governments to reimburse the debt. However, it should be noted that inflation (and hyperinflation) is costly for the economy, as seen in the previous problem set. Inflation reduces the value of fixed annuities (AVS, disability insurance annuities, income from fixed interest on savings, etc) which could unequally impact the wealth of different individuals in the society.

Exercise 5

Suppose that the Swiss government puts in place a fiscal policy whose effects are represented in the graph below. The level of production is expressed in billions of francs.



1. Discuss the graph (shift in the curve, change in the level of production and prices).
2. Knowing that the production function is given by $Y = 500,000L$ (where Y is the level of production and L is the number of workers) and that the active workforce is equal to 20 million people, calculate the natural unemployment level U_N , the unemployment level U and the inflation rate π .
3. Give the equation of the Phillips curve in the short run. Discuss, and then calculate the value of the slope. Draw the Phillips curve.
4. What is the level of the expected inflation at point A ? What would happen if the Swiss National Bank (SNB) were to act so as to maintain the inflation rate at the same level (as in point B) in the years to come? Show graphically (using a Phillips curve) what would happen. Discuss.

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1. A fiscal policy which shifts the AD curve to the right is an expansive fiscal policy. As a consequence of decreased taxes, the demand is stimulated and therefore production increases. Production passes from 8500 to 8600 billions of francs, and the prices increase from 100 to 105 francs.

2. Seeing as $Y = 500,000L$, we then know that for $Y = 8,500$ billion it must be that there are $L = 17$ million workers. We are also told that the active population is 20 million people. It follows then that the number of unemployment people is 3 million people. This represents 15% of the active population, and so we conclude that the natural level of unemployment $U_N = 15\%$.

Calculating the unemployment rate U at point B : We know that at point B the level of production is 8,600 billion. Using the production function, we can determine the number of workers in point B by dividing 8,600 billion by 500,000. We obtain 17,2 million workers. Seeing as the active population is 20 million people, we can deduce that the unemployment rate is 14%.

Calculating the inflation rate π at point B : We can observe that after the fiscal policy was introduced the prices have passed from 100 to 105. Hence, the inflation rate is $\pi = 5\%$.

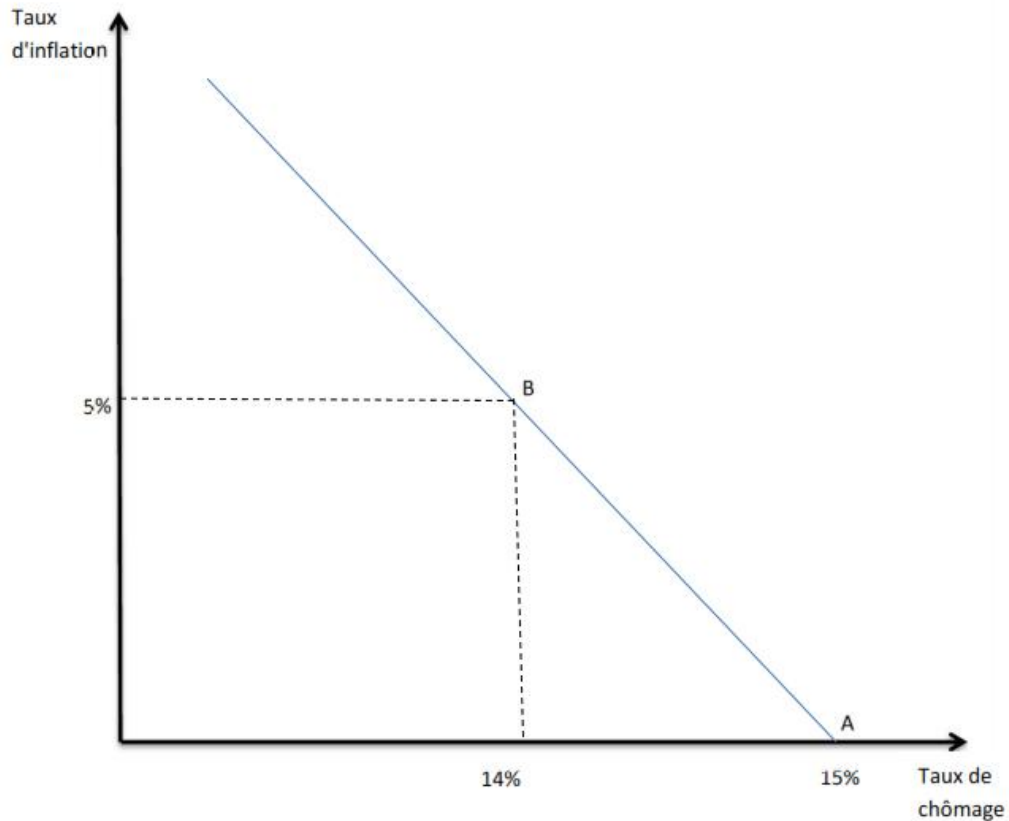
3. *Phillips curve equation in the short-run :* The Phillips curve corresponds to the following equality :

$$U = U_N - a(\pi - \pi^e) \quad (1)$$

This equation suggests a negative relationship between the inflation rate and the unemployment rate. As such, for any value of effective inflation rate, an increase in expected inflation should result in an increase in the unemployment rate.

Calculating the value of slope a : Starting from the equality $U = U_N - a(\pi - \pi^e)$ we can easily identify a . We know that the values of U , U_N , and π are respectively 14%, 15% and 5%. In addition we know that at the long term equilibrium the inflation expectation are perfectly rational and that the unemployment rate is at its natural level. Thus, in line with the Phillips curve we can deduce that the inflation is zero, and that therefore the expected inflation is zero as well. We have then $\pi^e = 0$. Seeing as the inflation expectations do not change in short term, we can deduce a from the Phillips curve as follows : $14\% = 15\% - a(5\% - 0\%)$. Therefore it must be that $a = 0.2$.

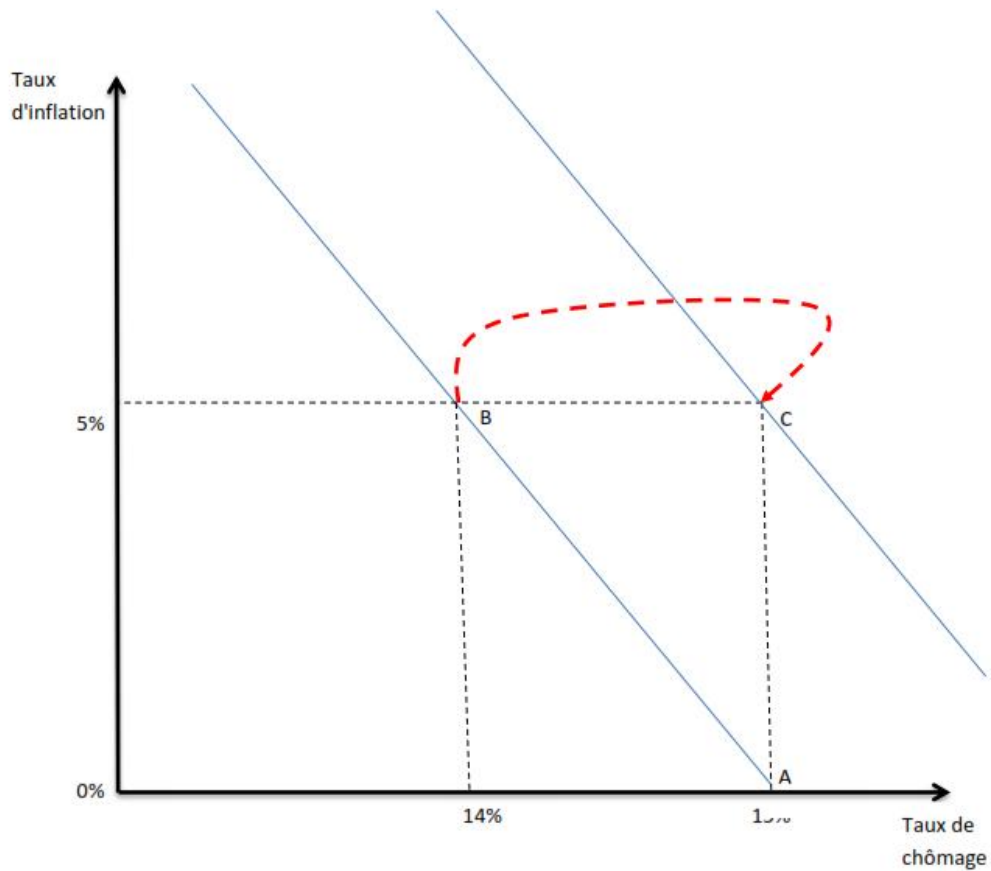
Graphic representation of the Phillips curve :



4. *Expected inflation rate at point A* : As we concluded above, the value of expected inflation is $\pi^e = 0$.

SNB maintains the inflation rate at same level as in point B : If the SNB was to choose to maintain the inflation level at 5% during the years to come, this would mean that the expected inflation would grow to same level., ie $\pi^e = 5\%$. The compromise would be at a disadvantage to the unemployment rate, which would pass to 15% in the long-term.

Graphic representation :



Explanation : The movement from point B to point C is represented by the red line. This movement represents the adjustment process of expected inflation. Seeing as expected inflation passes above 0%, the effective inflation rate increases above 5% and the unemployment increases as well. Seeing as the effective inflation is above 5%, the expected inflation will also grow beyond this value. In order for the π^e and π to come back to their long-term level (ie 5%), the unemployment rate would need to surpass 15% (its natural level). This is why the red line initially surpasses point C before eventually approaching the point C.